Predicting diabetes health outcomes using a variety of models

Project Category: Application

Name: David Wang
SUNet ID: davidw23
Department of Computer Science
Stanford University
davidw23@stanford.edu

1 Introduction

According to the CDC, nearly 1 in 10 Americans has diabetes, costing an estimated $327 billion dollars annually in healthcare cost. In fact, diabetes is the seventh leading cause of death in America. As such, gaining a better understanding of how to treat diabetes could lead to serious improvements in healthcare outcomes in general. One way to measure the effectiveness of treatment is patient readmission rate, whether or not a patient is readmitted to the hospital within a certain amount of time after being discharged. The goal of this project is to predict whether a given patient who received treatment for diabetes at a hospital will be readmitted to the hospital within 30 days. This project attempts to do so given some demographic information such as age and weight and more importantly information about the treatment the patient received during their hospital stay. In doing so, we can then attempt to answer the question of what treatments lead to better health outcomes by interpreting the resulting classifiers. By exploring various data processing methods and machine learning methods, we can try to gain insight into what variables are the best predictors of diabetes readmission.

2 Related Work

Previous machine learning work on diabetes tended to focus on the more basic question of whether or not we can predict serious hospitalizations due to diabetes. The question of what treatments were effective for current diabetes patients is not something that is explored in the literature. As such, while previous work can be a helpful reference for training the classifiers, the interpretation of this project's classifiers is novel.

One limitation common to almost all previous work is the size of the datasets. For instance, the Pima Indian Diabetes Dataset (PIDD) consists of only about 700 patients [1] but has been used repeatedly in the literature with some authors reporting accuracy rates as high as 90% [2]. While this is very promising, the small sample size of the dataset could mean that previous works are overfitting to the test set rather than actually evaluating the effectiveness of their classification methods. The largest dataset I came across during my literature review consisted of 40,000 patients from the Boston area, which reported an accuracy of 89% [3].

As can be expected, the techniques used across is very diverse. Another paper which uses the same PIDD dataset uses logistic regression, support vector machines, and random forests to achieve a respectable accuracy of around 84% [4]. Other work has tried using latent variable models like naive bayes to a high degree of success, again on the PIDD dataset [5]. The aforementioned team who worked on the dataset from Boston used a particularly interesting combination of clustering before using more traditional classification methods [3]. This work is particularly meaningful due to the much more statistically significant dataset they were working on. Another interesting work proposed an ensemble of pretty much every technique used in the literature. Rather than picking a single best classifier, the authors of this paper trained classifiers using kNN, Decision Trees, Random Forest, Naive Bayes, and MLP, then took the weighted average of the predictions [6]. To calculate the weights they used another performance metric common in the literature which is the area under curve (AUC) of the Receiving Operating Characteristic (ROC) curve. This curve represents the tradeoff between true positive rate and false positive rate. Intuitively, better classifiers will have sharper tradeoffs and thus larger AUC, meaning we should weight these classifiers more heavily in the resulting ensemble.

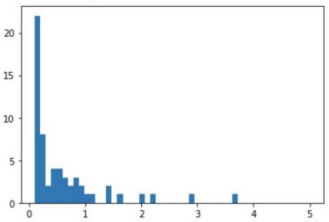
3 Dataset and Features

For my dataset I chose to use a publicly available dataset that was created by researchers in 2013 [7]. It consists of over 100,000 datapoints in total and each datapoint contains 47 features, with a couple of caveats. Since there is no pre-existing train test split, I randomly created a 80 % train/val 20% test split which I kept consistent across all experiments. A typical datapoint consists of several parts. First, a couple of identifying features including patient number, encounter id, and payer code. Then, demographic information such as race, gender, age (in 10 year ranges), and weight. Third, general information about their particular hospital visit such as admission source (transfer, referral etc.), admission type (emergency, trauma etc.), discharge disposition (discharged to home, transferred, etc.) as well as number of procedures, number of medications, number of days spent in hospital, number of diagnoses made by physicians and the specific diagnoses of medical ailment made according to the ICD 9 standard. Most importantly, a list of possible drugs and whether or not they were administered

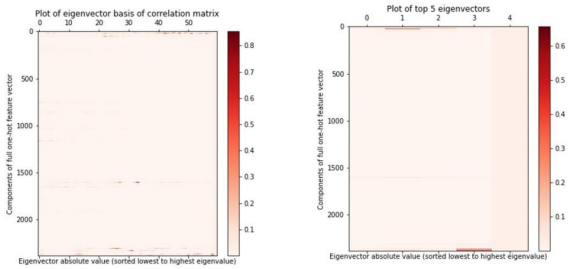
to the patient, and a label for whether they were readmitted within 30 days, after 30 days, or not at all.

There are a couple of problems with the dataset that need to be addressed first before using for classification. First of all, it is clear that the columns of the dataset cannot be represented with simple binary features. We could take only the ordinal number features and hope for the best, but this would mean that we lose out on much of the information left in diagnosis type, demographic information, discharge type etc and all other information which is nominal rather than ordinal. Similarly, the labels themselves are not simple binary labels, rather they are labels for a multi-class classification problem and thus need to be converted into one-hot labels, or converted into a binary label by grouping the < 30 and > 30 classes into a single "readmitted" class. Both approaches for both the dataset and the labels will be explored. The other problem is that many features are missing a large amount of values. For instance, weight is missing nearly 98% of their values. Other features are missing only a small amount, 1 or 2 %. Since all variables are either missing more than 50% of their values or less than 2%, for those that are missing more than 50% we remove the column altogether and for those less than 2% we simply ignore those datapoints when they appear. Lastly, features which should not be taken into account or provide no information such as patient ID or 'citoglipton' which contains only a single value can also be dropped.

In order to address the nominal features, we could take every nominal feature, turn it into a one-hot vector and then simply stack all of the one-hot vectors with the ordinal features into one giant feature vector. However, due to features such as the ISP-9 diagnosis which can take on several hundred unique values in this dataset, the dimensionality of such a feature vector would be greater than 2000. Thus, after encoding the data into stacked one-hot encoders it is necessary to attempt to reduce the dimensionality by using Principle Component Analysis (PCA). After computing the eigenvalues of the correlation matrix I plotted all eigenvectors with eigenvalues which that were above 0.1 which yielded a reasonable 60 features, shown below.



To visualize which features were important and attempt to find patterns, I also plotted the resultant top 5 eigenvectors and the entire eigenvector basis.



. The top eigenvector's absolute value, plotted as number 4 above is pretty much uniform. This is to be expected because we are encoding a mostly one-hot vector which contains a large amount of zeros, and after standardizing this becomes a bunch of +/- 1. As such the principle component will consist of up to a constant factor mostly +/- 1. To get a better sense of what each component plotted above corresponds to I printed the most important components of each eigenvector in the top 5.

```
vector number 1:
------------------------------
time_in_hospital with val -0.13943541357654413
num_medications with val -0.23077410781429242
metformin_No with val 0.14181043810485042
metformin_Steady with val -0.12672687451885553
insulin_Down with val -0.12396684692417416
insulin_No with val 0.3799918536214074
insulin_Steady with val -0.13786387809307118
insulin_Up with val -0.11759804268172654
change_Ch with val -0.45021923375116374
change_No with val 0.4502192337511638
diabetesMed_No with val 0.34470569287897596
diabetesMed_Yes with val -0.344705692878976
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As expected, insulin and diabetes medication are highly representative of the dataset, as well as

time in hospital and number of medications. However, qualitative analysis of my other principle components suggest another potential shortfall of the dataset. The components which best describe the correlation matrix of the dataset tend to have more to do with demographic information than actual treatments. As such the classifiers may simply learn to predict whether people are more or less at risk of severe complications rather than what treatments are more or less helpful. Even when the components do involve information about treatment such as the "Change" component which represents change in dosage of medication, this can never provide enough information to understand what about the treatment was harmful or beneficial. Nevertheless, we can hope that our classifier performs well enough that it can separate points with similar demographics but different treatments such that it does learn the desired information about effectiveness of different treatments.

4 Methods

All classification was done using the extensive Scikit-Learn libraries which made comparison between all many classifiers I used in this project far less time consuming [8]. The first classifier that I used was simply kNN. I evaluated this both on only the ordinal number features, and the newly created PCA features as a proof of concept.
The next classifier was logistic regression. Since I had to adapt this to multi-class classification for this problem I used the generalization from the bernoulli distribution to the multinomial distribution. This generalization of the exponential family distribution the model is based on means the hypothesis is instead given by the softmax and the loss is given by the cross-entropy loss.

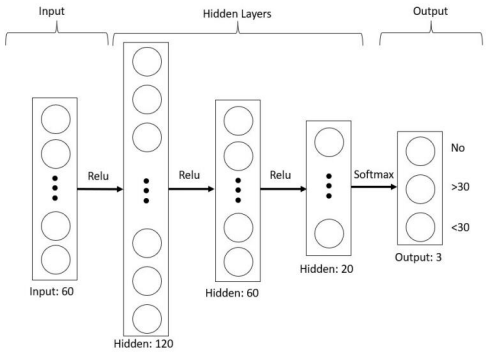
$$L = -\frac{1}{N}\sum_{i=1}^{N} y_i \log(\hat{y}_i)$$

where $\hat{y}_i$ is the estimated probability for that label.

The next class of models I experimented on were Decision trees which require regularization by setting a maximum depth on the learned tree. This prevents the decision tree classifier from ever getting close to fully learning to distinguish every point in the dataset uniquely and thus allows it to generalize. I experimented with different maximum depths to find the optimal classifier, which I then evaluated using the test set.

As a natural continuation to decision trees, I then experimented on Random Forests. These can be thought of as an alternative method of regularization decision trees. Rather than allowing the decision tree to split nodes using all possible features, random forests only allow a randomly selected few features, for my experiments only 7 out of 60. In order to ensure that all the information from the dataset is included, random forests instead average over a large ensemble of these smaller trees which I chose to contain 100 random trees. Again, I experimented with maximum depth and compared the results between the random forest and decision tree.

Next, I tried implementing a neural network (or multi-layer perceptron), the exact architecture of which is shown below:

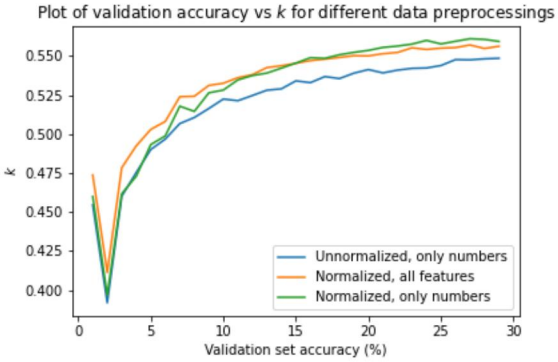


I chose the hidden layers somewhat arbitrarily with some minor tweaking until I found a neural network which could easily overfit on the train set. Then I used early-stopping by splitting the train into train set and val set to stop when the val set accuracy no longer improved. The loss function for this is again the cross-entropy loss. I tuned the learning rate hyperparameter to find the optimal neural network which I then evaluated on the test set.

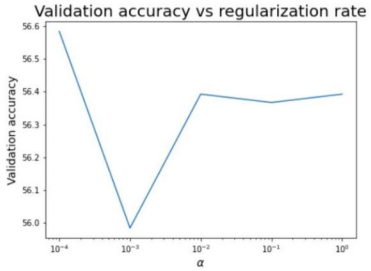
Lastly, one more experiment was done using the aggregated model as described previously in the literature review. Since it was easiest to calculate AUC for binary classification, this was only done on binary labels.

5 Results and discussion

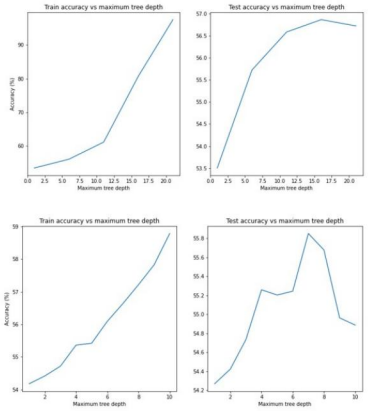
As a first experiment, I compared the results of the kNN classifier on data that included only the numeric features against the results on data that included all the features after PCA. As expected, both classifiers performed better with the full features, normalization of the data further improved the accuracy. Using this as a proof of concept, in all future experiments I used all of the features, after processing with PCA. The results of the experiment are shown below:



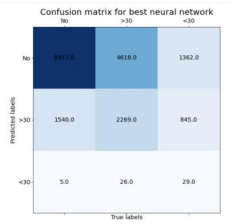
For the neural network I tuned the regularization hyperparameter and obtained the following graph suggesting the optimal L2 regularization rate was 1e-4.



Lastly, below are the results for the optimal maximum depth of both the decision tree classifier and the random forest classifier. As expected, we see that since the random forest classifier already has much stronger regularization built in, it is able to reach larger maximum depths. Moreover, we see in both the expected behavior of overfitting on the training set leading to decreased validation accuracy.



In order to get a sense of the classification boundaries compared between all the classifiers, I plotted confusion matrices for each and calculated the test accuracy for the best classifier of each type. Below is just an example for the neural network.



In general, I found that all classifiers tend to confuse no readmission and readmission after 30 days the most. Moreover, it is quite apparent that while the classifiers did confuse readmission within 30 days less, this could be due to the fact that such datapoints are far rarer in the dataset in general and thus there are less opportunities to make mistakes. If we weight the accuracies between classes equally, we see that the accuracies across the board are much lower, although the relative accuracies do not change much.

Accuracies for multi-class labels

```
kNN achieved test set accuracy of: 0.38945818235753765
Logistic regression achieved test set accuracy of: 0.38536526409138966
Decision tree achieved test set accuracy of: 0.3809276320827294 5
Random forest achieved test set accuracy of: 0.38886082097946256
Deep neural network achieved test set accuracy of: 0.39914393163818235
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Accuracies for binary labels

```
Logistic regression achieved test set accuracy of: 0.6160828106674825
kNN achieved test set accuracy of: 0.6051705675386263
Decision tree achieved test set accuracy of: 0.59425832440977
Random forest achieved test set accuracy of: 0.6197032277803274
Deep neural network achieved test set accuracy of: 0.6270460455866605
Aggregate model achieved test set accuracy of: 0.6180714904900311
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6 Conclusion/Future Work

In conclusion, the deep neural network approach performed the best of all the machine learning techniques used in this project. However, the improvements between various algorithms were almost negligible. Even a comparatively naive approach such as kNN was almost able to match the performance of the best neural network in both the binary classification problem and multi-class classification problem. Moreover, with the exception of high bias classifiers like logistic regression, all classifiers were able to seriously overfit to training data and yet still performed very poorly on the test data which is taken from the same distribution. This is likely due to limitations of the dataset as pointed out in the methodologies section. Another serious problem encountered in this analysis was the huge disparities in accuracies between the different classes. Since the dataset consisted mostly of patients who were not readmitted, the accuracy was overinflated because all classifiers tended to predict no readmission far more than the other two classes. After averaging the accuracy between classes the best accuracy was only 38%, too low for meaningful predictions. Given more time it would also have been interesting to explore how training on the balanced dataset would affect overall accuracy and individual group accuracies achieved by the classifiers. In addition, while for this project I discarded latent variables, previous work on diabetes classification attempted to fill in missing values or used latent variable models such as naive bayes to high degrees of success. Although for this particular dataset filling in missing values could only have improved accuracy by a maximum of 2% it would have been interesting to see if a reasonable increase in accuracy could be achieved.

Ultimately, although this project was able to achieve test prediction accuracies above random chance this project was unsuccessful at providing conclusive interpretable results as to which factors are most important in determining diabetes patient health outcomes which was the initial motivation behind the stated goal of this project. For future work it would have been helpful run the random forest classifier on the data prior to using PCA, which would have made it easier to interpret the importance of each component of data. It may also have been fruitful to attempt to interpret the regular decision tree and the logistic regression model. In addition, it would have been useful to perform the same analysis done in this report on other available diabetes datasets in order to confirm that the poor classification accuracy was a consequence of the dataset, and not a consequence of an incorrect approach.

7 Contributions

I did everything by myself without outside help.

References

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